StayWise

Airbnb Market Intelligence Dashboard

Sector: Consumer Discretionary — Travel & Hospitality

Team EG-7 | Newton School of Technology
Akshat Agrawal • Pranjal Shukla • Sahil Chand • Shyam Verma • Trishit Swarnakar • Vikrant Yadav
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⬡ https://github.com/sahil-chand-1911/E_G-7_StayWise

⬡https://public.tableau.com/app/profile/akshat.agrawal2260/viz/Airbnb_17773079392460/MarketIntelligence

Submission: 29th April 2026

Context & Problem Statement

Sector Context

Travel & Hospitality is a cornerstone of Consumer Discretionary spending. Europe is one of Airbnb's most active markets with millions of listings. Post-pandemic recovery has intensified competition among hosts.

Who Decides?

Primary stakeholders are Airbnb hosts from single-property owners to professional managers overseeing multiple listings who need timely market intelligence to price, invest, and compete.

The Core Problem

94,000+ European listings. Zero actionable intelligence. Hosts misprice listings, over-invest in low-impact amenities, and lose competitive edge, all due to operating on intuition rather than data.

Objective: Develop a comprehensive market intelligence dashboard identifying key drivers of revenue and occupancy for European Airbnb listings enabling data-driven decisions on pricing, amenity investment, and market positioning.

Data Engineering — Source to Pipeline

Data Source

Dataset:	**Airbnb Market Data — Europe**
Source:	**Kaggle — Jason Airroi**
Listings:	**95,898 rows × 61 columns**
Past Rates:	**1,115,174 rows × 17 columns**
Period:	**Trailing Twelve Months (TTM)**

ETL Pipeline (5 Notebooks)

```
01_loading      → Ingest CSV, handle mixed dtypes,
                pickle raw data

02_cleaning     → Fill nulls, remove 570 zero-revenue listings,
                parse amenities
03_eda          → Distributions, segment comparisons, trend
                charts
04_statistical  → Amenity impact, correlation matrices,
                benchmarking
05_final_load   → Revenue tiers, market aggregations, export
                CSVs
```

Key Cleaning Steps

570 zero-revenue records removed (inactive listings)

14 mixed-type columns manually converted

Occupancy > 1.0 clipped; amenities < 5% frequency excluded

Key Columns:

`listing_id` | `city` | `listing_type` | `room_type` | `amenities` | `ttm_revenue` | `ttm_occupancy` | `rating_overall` | `cleaning_fee` | `latitude/longitude`

KPI & Metrics Framework

TTM Revenue

Sum of revenue over trailing 12 months (USD)

Core measure of listing profitability

04_statistical.ipynb

TTM Occupancy

Reserved days ÷ Total available days

Indicates demand & pricing competitiveness

03_eda.ipynb

RevPAR

Total Revenue ÷ Total Available Room-Nights

Industry-standard cross-market efficiency metric

05_final_load.ipynb

Avg Daily Rate

Total Revenue ÷ Total Reserved Days

Pricing position vs. competitors

04_statistical.ipynb

Revenue Tier

Quartile → Bronze / Silver / Gold / Platinum

Benchmarking against market segments

05_final_load.ipynb

Rev Perf Index

Listing TTM Revenue ÷ Market Mean Revenue

Competitive positioning signal

05_final_load.ipynb

Key Insights — Exploratory Data Analysis

01

Villas & Entire Homes Win

Entire homes and villas consistently occupy Gold & Platinum revenue tiers. Property type is the single strongest predictor of revenue tier — private rooms trail far behind.

02

Coastal vs. Inland Gap

Bodrum delivers 2–3× revenue per unit vs. inland urban markets like Istanbul. Lower listing density × higher nightly rates = outsized per-unit profitability.

03

Average 34.6 Amenities

WiFi, A/C, kitchen appear in 80%+ of listings. Premium amenities (pool, sea view) are in only 5–10%, creating a scarcity premium that directly elevates revenue.

04

Sharp Summer Seasonality

Historical rates show a pronounced summer peak in both occupancy and daily rates. Off-peak months see steep declines — dynamic pricing is massively under-utilised.

05

Revenue Is Right-Skewed

Top quartile (Platinum) earns disproportionately more than the bottom three. Silver → Gold gap is narrower than Gold → Platinum, making targeted upgrades high-ROI.

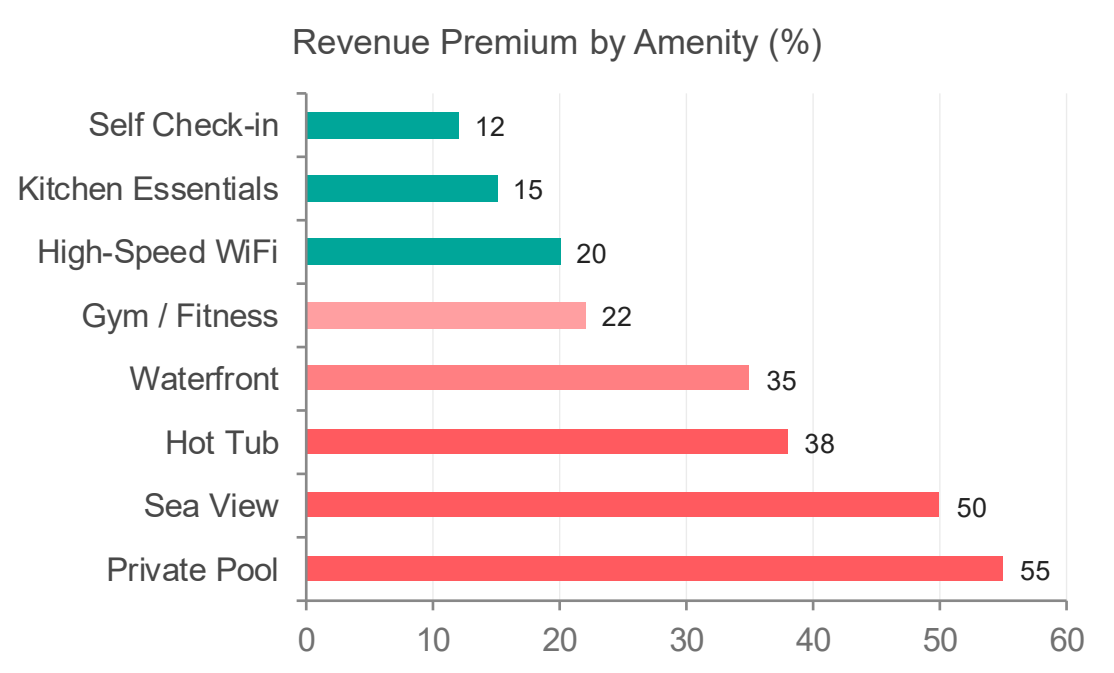
06

Amenity Count Signals Performance

Listings with 30+ amenities consistently outperform those with fewer. Completeness of listing inventory correlates strongly with occupancy tier.

Source: notebooks/03_eda.ipynb — All charts committed to GitHub repository

Advanced Statistical Analysis



Revenue Multipliers

Private pool, sea view, hot tub → 40–60% above average
Justify higher nightly rates. Scarcity creates premium demand.

Occupancy Anchors

WiFi, kitchen essentials, self check-in → 15–25% off-peak occupancy lift
Serve business & extended-stay guests year-round.

Methodology: Amenity explosion → group mean revenue/occupancy → filter to 5%+ frequency (91 amenities) → ranked by performance | Libraries: Pandas, NumPy, Scipy, Seaborn | Source: 04_statistical_analysis.ipynb

Tableau Dashboard Walkthrough

Executive View (CEO)

- High-level KPI summary (TTM Revenue, Occupancy, RevPAR)
- Geographic distribution of listings across European markets
- Revenue tier breakdown: Bronze → Silver → Gold → Platinum
- Market Intelligence overview with city-level comparison
- Top amenity categories driving revenue premium

Operational View (Manager)

- Pricing & performance trends by city and property type
- Amenity gap analysis: identify missing revenue-driving features
- Revenue Performance Index vs. market benchmarks
- Seasonal occupancy trends for dynamic pricing calibration
- Drill-down filters: city, property type, revenue tier, date

Interactive Filters: City | Property Type | Date Range | Revenue Tier

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Recommendations

R1	*Insight: Pool, sea view & hot tub → 40–60% revenue premium* **Invest in Premium Amenities**	📊 +40–60% annual revenue | ROI in 12–18 months | Budget: $5K–$20K	Medium
R2	*Insight: WiFi & kitchen = top occupancy anchors, esp. off-peak* **Perfect Work-From-Home Features**	📊 +15–25% off-peak occupancy | Cost: $500–$2,000 | Implementable in days	High
R3	*Insight: Coastal cities deliver 2–3× revenue vs. inland urban* **Focus Acquisition on Coastal Markets**	📊 +100–200% portfolio revenue per unit vs. inland equivalents	Medium
R4	*Insight: Sharp seasonal patterns — flat pricing leaves money on table* **Implement Dynamic Pricing**	📊 +10–20% annual revenue | No capital cost | Native in Airbnb tools	High
R5	*Insight: Hosts lack actionable benchmarking intelligence* **Embed Dashboard in Monthly Operations**	📊 Compounding KPI improvement through evidence-based iteration	Immediate

Impact & Business Value

+40–60%
Revenue lift
from premium amenities

+15–25%
Off-peak occupancy
from WFH upgrades

2–3×
Portfolio revenue
coastal vs. inland

+10–20%
Annual revenue
from dynamic pricing

Implementation Timeline & Expected Impact

Week 1–4
WFH upgrades + dynamic pricing

Month 1–3
Dashboard adoption + pricing calibration

Month 3–6
Amenity gap identification + planning

Month 12–18
Premium amenity ROI achieved

⚠ Cost of Inaction: Professional property managers are deploying data-driven strategies at scale. Hosts who don't adopt market intelligence risk being systematically undercut on pricing and outcompeted on amenity quality.

Limitations & Next Steps

Limitations

Geographic Scope

Strong Turkey/Eastern Europe bias — findings may not fully generalise to Western European markets.

Single TTM Snapshot

One trailing year of data; post-pandemic structural changes may not be fully captured.

Selection Bias

Dataset likely over-represents active, successful listings. Inactive hosts may behave differently.

Correlation, Not Causation

Amenity impact analysis identifies associations. High-revenue listings may have pools because they are already premium properties.

Next Steps

Predictive Revenue Model

Random Forest / XGBoost to forecast TTM revenue pre-investment — enabling ROI estimation before spending.

Dynamic Pricing Engine

Algorithmic nightly rate adjustment based on demand signals, competitor ADR, and seasonal patterns.

Guest Sentiment Analysis

NLP on review text to identify the specific experience factors driving 5-star ratings and repeat bookings.

Real-Time Dashboard

Live Airbnb API feed into Tableau for up-to-the-minute market intelligence and competitor monitoring.

StayWise — Turning Airbnb Data Into Competitive Advantage for European Hosts